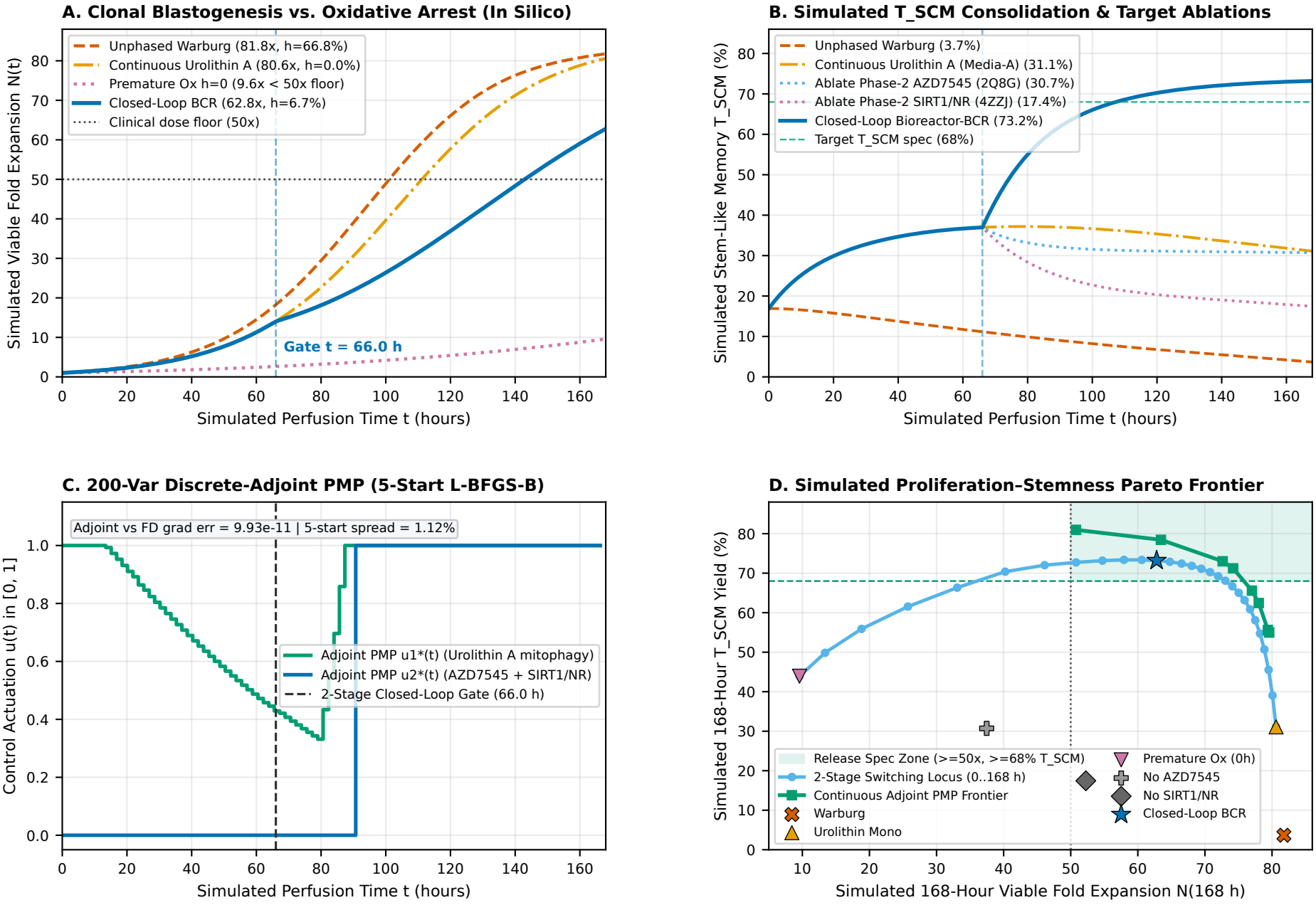
A. Clonal Blastogenesis vs. Oxidative Arrest (In Silico)
Unphased Warburg (81.8x, h=66.8%)
Continuous Urolithin A (80.6x, h=0.0%)
Premature Ox h=0 (9.6x < 50x floor)
Closed-Loop BCR (62.8x, h=6.7%)
Clinical dose floor (50x)
Gate t = 66.0 h
Simulated Viable Fold Expansion N(t)
Simulated Perfusion Time t (hours)
B. Simulated T_SCM Consolidation & Target Ablations
Unphased Warburg (3.7%)
Continuous Urolithin A (Media-A) (31.1%)
Ablate Phase-2 AZD7545 (2Q8G) (30.7%)
Ablate Phase-2 SIRT1/NR (4ZZJ) (17.4%)
Closed-Loop Bioreactor-BCR (73.2%)
Target T_SCM spec (68%)
Simulated Stem-Like Memory T_SCM (%)
Simulated Perfusion Time t (hours)
C. 200-Var Discrete-Adjoint PMP (5-Start L-BFGS-B)
Adjoint vs FD grad err = 9.93e-11 | 5-start spread = 1.12%
Adjoint PMP u1*(t) (Urolithin A mitophagy)
Adjoint PMP u2*(t) (AZD7545 + SIRT1/NR)
2-Stage Closed-Loop Gate (66.0 h)
Control Actuation u(t) in [0, 1]
Simulated Perfusion Time t (hours)
D. Simulated Proliferation–Stemness Pareto Frontier
Release Spec Zone (>=50x, >=68% T_SCM)
2-Stage Switching Locus (0..168 h)
Continuous Adjoint PMP Frontier
Warburg
Urolithin Mono
Premature Ox (0h)
No AZD7545
No SIRT1/NR
Closed-Loop BCR
Simulated 168-Hour T_SCM Yield (%)
Simulated 168-Hour Viable Fold Expansion N(168 h)